# THE DEVELOPMENT OF BLUETOOTH LOW ENERGY IN CISCO WLAN

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**IN COMPUTER ENGINEERING**

**MAE FAH LUANG UNIVERSITY**

**2022**

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A SENIOR PROJECT SUBMITTED TO
MAE FAH LUANG UNIVERSITY IN PARTIAL FULFILLMENT OF
THE REQUIREMENTS FOR THE DEGREE OF
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(Aj. Dr.Surapol Vorapatratorn)

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( Asst. Prof. Dr. Suppakarn Chansareewittaya)

# ACKNOWLEDGEMENT

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**Degree** Bachelor of Engineering
(Computer Engineering)

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| | Aj. Dr.Surapol Vorapatratorn | Committee |
| | Aj. Suppakarn Chansareewittaya | Committee |

**Abstract**

This senior project proposed the development of a BLE (Bluetooth Low Energy) that is a IoT(Internet of Things) for tracking the durable articles and display the result through website called "MFU AP Mmap".Use python to develop the website. This senior project will help crew to find durable articles easily.

**Keyword:** Bluetooth Low Energy (BLE), Tracking, Python, Internet of Things (IoT

# CONTENTS

## LIST OF TABLES



## LIST OF FIGURES

Figure 3.3.3 All tracking

# CHAPTER 1
# INTRODUCTION

## 1.1 Background and Rational

Mae Fah Luang undergraduate is faced with the problem of missing durable articles or wrong location, even if the durable articles number is clearly stated. This problem still exists. Therefore, we invented the innovation of the Internet of Things or we called “IoT”, using BLE (Bluetooth Low Energy) as the signal transmission for device tracking. We will use Python language to display the tracking results on the MFU AP map.

## 1.2 Objective

This project was created to track the durable articles in Mae Fah Luang University.

## 1.3 Scope

We can use BLE to tracking the durable articles in Mae Fah Luang University.

## 1.4 Methodology

1.4.1 Design physical system

1.4.2 System Identify

1.4.3 Process the system

1.4.4 Design the control algorithm

1.4.5 Test the system

1.4.6 Final Presentation

## 1.5 Plan

Pre-project Project

**Table 1.5.1** Working plan

| | Senior project 1 | | | | | Senior project 2 | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Plan/Month** | Jan | Feb | Mar | Apr | May | Aug | Sep | Oct | Nov | Dec |
| Proposal preparation and defense | | | | | | | | | | |
| Literature review and requirement gathering | | | | | | | | | | |
| System analysis | | | | | | | | | | |
| System design | | | | | | | | | | |
| Senior project 1 document preparation | | | | | | | | | | |
| Senior Project progress1 | | | | | | | | | | |
| System development | | | | | | | | | | |
| System testing | | | | | | | | | | |
| Senior project 2 document preparation | | | | | | | | | | |
| Final Presentation | | | | | | | | | | |

## 1.6 Expected Result

Everyone can use our BLE for tracking to save more time that crew have to find out where durable articles are.

## 1.7 Place and Equipment

1.7.1 Mae Fah Luang University

1.7.2 Hardware (BLE,ESP32)

1.7.3 Software (Python,Arduino)

# CHAPTER 2

# LITERATURE REVIEW

## 2.1 Related Theory

### 2.1.1 DNA Space

Cisco DNA Spaces is a location-based services platform developed by Cisco that uses Wi-Fi and other location technologies to provide real-time insights and analytics for businesses and organizations.

Cisco DNA Spaces is designed to help organizations understand how people move and behave within physical spaces such as buildings, campuses, and stores. The platform can collect data from various sources, including Wi-Fi access points, Bluetooth beacons, and other location sensors, and use this data to generate insights and analytics that can help organizations optimize their operations, improve customer experiences, and increase revenue.

### 2.1.2 Bluetooth Low Energy (BLE)

Also known as Bluetooth Smart, is a wireless technology designed for low-power, low-latency communication between devices. BLE is a variant of the Bluetooth wireless technology, and it was introduced in Bluetooth 4.0.

BLE is designed to enable wireless communication between small devices with low power consumption requirements, such as sensors, wearables, and other IoT devices. BLE is well-suited for these types of devices because it uses significantly less power than traditional Bluetooth technology, which can help to extend battery life.

One of the key features of BLE is its ability to operate in two different modes: peripheral and central. In peripheral mode, the BLE device broadcasts its presence and data to other devices in the vicinity, while in central mode, the BLE device connects to and receives data from peripheral devices.

BLE also includes a number of security features to protect against unauthorized access and data theft. These features include encryption and authentication protocols, as well as the ability to set access control policies and manage device permissions.

BLE is supported by a wide range of devices, including smartphones, tablets, and other computing devices. This makes it a popular choice for developers looking to create new IoT devices and applications that require low-power wireless connectivity.

Figure 2.1.2 Bluetooth Low Energy (BLE)

### 2.1.3 Web Application

A web application is a computer program that uses a web browser to perform a particular function. It is also called a web app. Web apps are present on many websites. A simple example is a contact form on a website.

A web application is a client-server program. It means that it has a client-side and a server-side. The term "client" here refers to the program the individual uses to run the application. It is part of the client-server environment, where many computers share information. For example, in the case of a database, the client is the program through which the user enters data. The server is the application that stores the information.

### 2.1.4 Python Object Oriented Programing

Structured computer programming has a separate character of the program in data management By using the form of a function or a subprogram, there are disadvantages to functioning. for use in the development of more complex programs, when applied, the program has along with the complexity Therefore, there is a problem of bringing to develop, improve, change and fix. which will be very difficult Therefore, a concept called “Object Oriented Programming” was developed.

### 2.1.5 Postman

Postman is an API platform for building and using APIs. Postman simplifies each step of the API lifecycle and streamlines collaboration so you can create better APIs-faster.

Solve problems and support complex program development, this making it popular to be used for development.many at present.

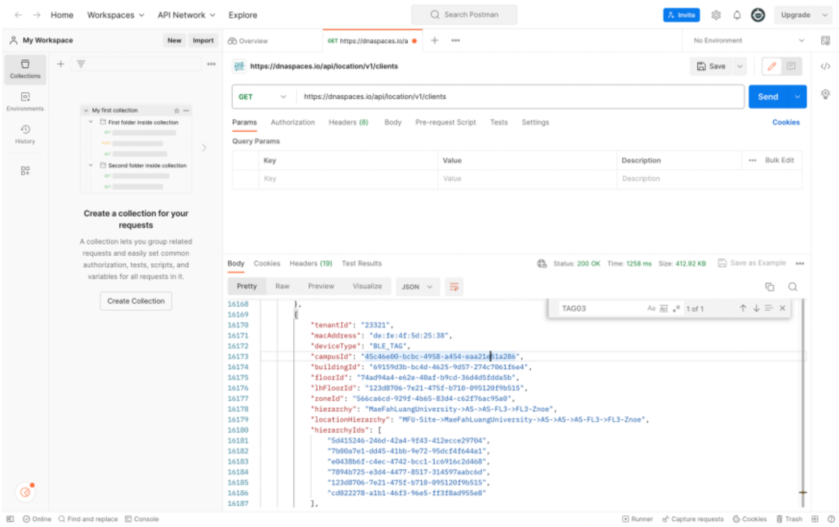


Figure 2.1.5 Postman

## 2.2 Related Work

### 2.2.1 BLE beacons

Beacons are transferred via the Bluetooth Low Energy standard – BLE for short – already mentioned. With this radio technology, multiple devices can be connected together. The advantage of BLE over other radio technologies is that BLE can provide low power consumption and low cost. That is why BLE is primarily used for mobile devices such as smartphones or smart watches. iOS supports this standard from iOS7, Android from version 4.3 and Windows Phone from Windows Phone 8. However, there is currently a problem with Android devices: not all Android devices with at least Android version 4.3 installed can communicate with beacons. It is possible that the Android device's hardware has BLE capability, but the function is still not used due to the lack of ROM updates. Beacons can be connected to devices, tools, and objects as Bluetooth beacons, which are small, rugged radio transmitters (Ø3.5-5.5 cm). Beacons use versions of the Bluetooth 4.0 standard. : Bluetooth low energy beacons (BLE beacons) with long lifespan up to 15 years. The signal range is about 50-100 m transmitted to the environment uniformly while identifying each device.

## 2.3 Related Technology

Identify both hardware and software required for the project as well as their references. The details should include the target version and you could explain each item separately as a new subtopic.

Kinect is a motion-detection camera developed by Microsoft Company. It is designed to let users interact with a videogame player or a computer through body gesture or voice.

# CHAPTER 3

# PROCESS AND DESIGN

## 3.1 Process

In pursuit of optimizing our ongoing project's development, a pivotal meeting was convened with the eminent technology company, Cisco. The primary objective of this meeting was to present our project's vision and seek valuable consultation from industry experts. Cisco, renowned for its cutting-edge innovations, demonstrated unwavering support by providing us with a trial license, which granted us a 120-day opportunity to leverage their resources. This essay delves into the key milestones achieved during our collaboration with Cisco, focusing on aspects such as API integration, web design using Figma, Python code development, and BLE device tracking for automatic positioning.

To establish effective communication between our project and the dnaspaces.io web server, the first step entailed obtaining an API KEY from dnaspaces.io. This vital key facilitated a secure and seamless connection, enabling us to harness the potential of dnaspaces.io's extensive capabilities. Rigorous testing through the Postman tool was conducted to ensure the functionality and reliability of the acquired API KEY. Such thorough evaluation bolstered our confidence in the robustness of our integrated solution, laying the groundwork for a successful project development journey.

With the fundamental infrastructure in place, we turned our attention to crafting an engaging and user-friendly web platform. Figma, an industry-leading design tool, proved instrumental in this endeavor. The platform allowed us to meticulously design the user interface, taking into account the project's specific requirements and intended user experience. The result was a visually appealing and intuitive web design that would serve as the foundation for our project's front-end.

Harnessing Python's versatility, we began developing a codebase to test and display MAC addresses, a critical aspect of our project. The Python code was meticulously designed to present the information in a structured tabular format, precisely tailored to the asset points within the university. By skillfully managing data presentation, we ensured that the relevant information was readily accessible, empowering project stakeholders with meaningful insights.

With the foundational aspects in place, we embarked on an ambitious endeavor to enhance the project's functionality further. Leveraging Bluetooth Low Energy (BLE) technology, we developed sophisticated Python code to track and monitor BLE devices' movements. The code dynamically identified the devices' positions, offering real-time information with unmatched precision. This automatic positioning feature significantly bolstered the project's practicality and potential applications, positioning us on the forefront of innovation.

The collaboration with Cisco proved to be a transformative experience, propelling our project's development to new heights. Through API integration, Figma-driven web design, and Python code development, we constructed a formidable technological framework. The culmination of these efforts was evident in the seamless display of MAC addresses and the revolutionary BLE device tracking system. As we continue to refine and expand our project, the support from Cisco serves as a testament to the power of collaborative efforts in harnessing technology for efficient and impactful solutions. Together, we embrace the future with the promise of innovation and groundbreaking achievements.

### 3.2 Activity diagram

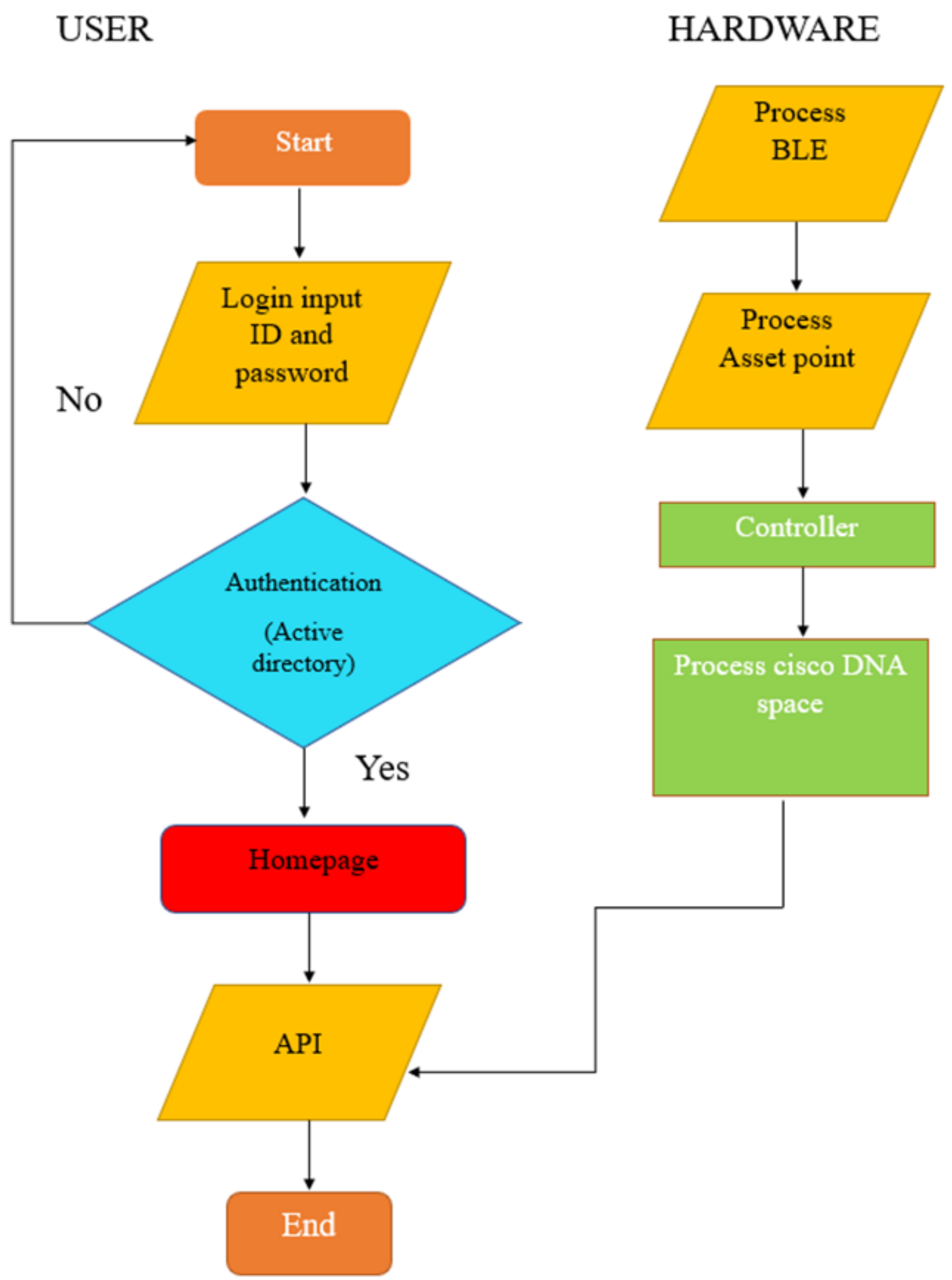


Figure 3.2 Activity diagram

## 3.3 Web design

### 3.3.1 Homepage

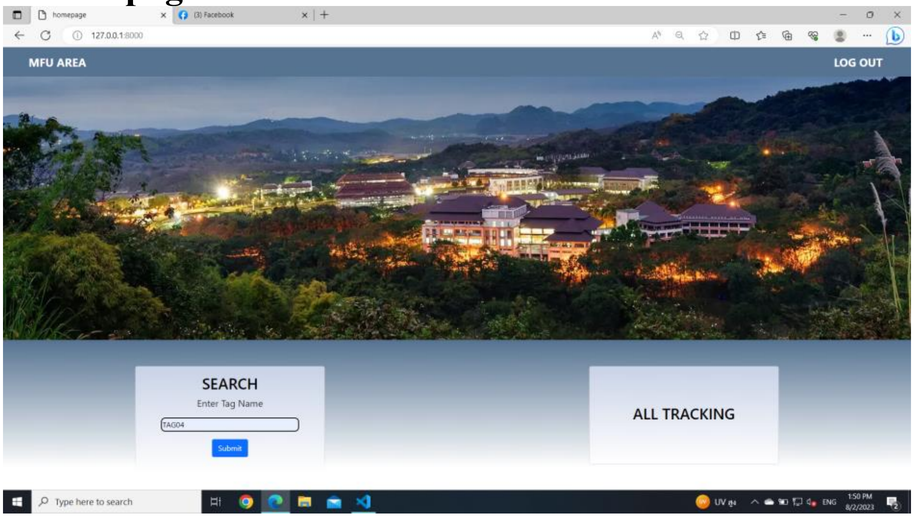


Figure 3.3.1 Homepage

### 3.3.2 Search tag results

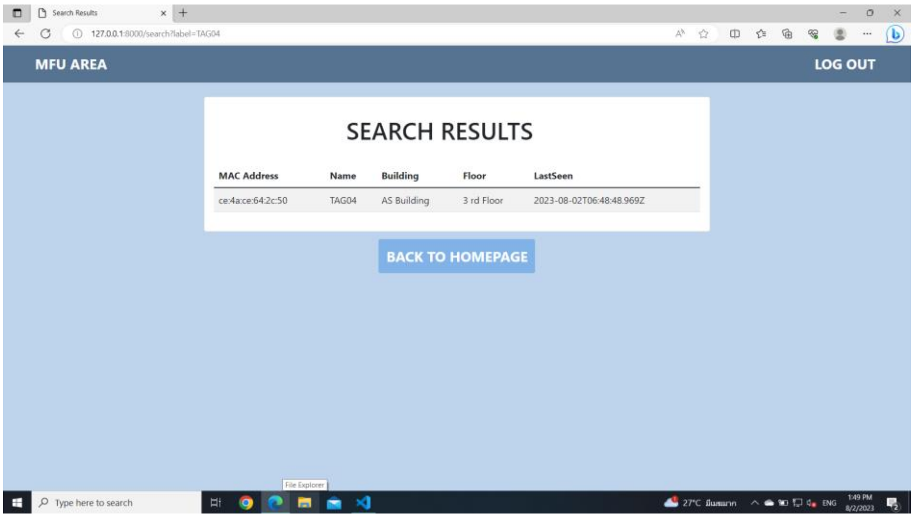


Figure 3.3.2 Search tag results

## 3.3.3 All tracking

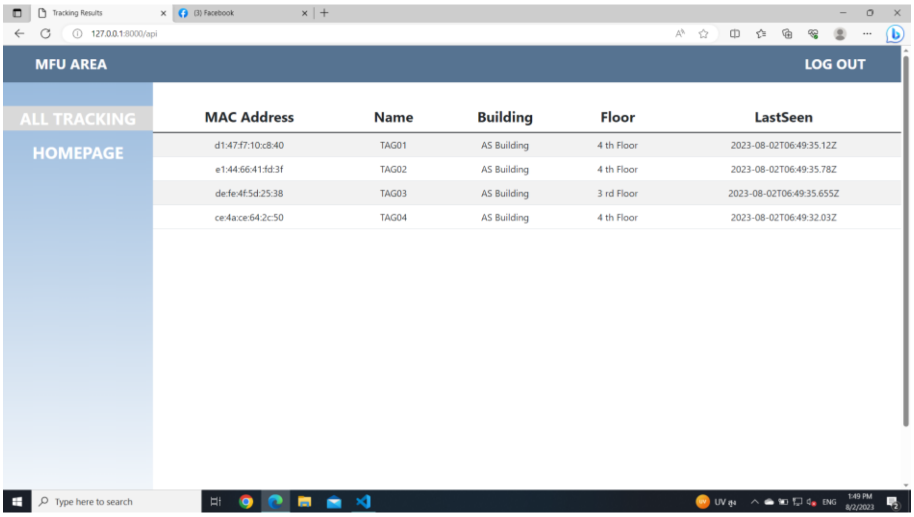


Figure 3.3.3 All tracking

# CHAPTER 4

# CONCLUSION

## 4.1 Conclusion

This project introduces a groundbreaking solution to address the challenges of tracking durable articles in stalls at Mae Fah Luang University. Through the development of a web-based platform, the system can efficiently identify and locate sought-after items, streamlining the process and benefiting both employees and customers.

Moreover, the proposed system's adaptability enables its application in various buildings, including educational institutions, large shopping malls, and facilities with dispersed areas such as shops, toilets, and ATMs. By utilizing Bluetooth technology, specifically the DNA Spaces system renowned for its stability and resolution, the system ensures reliable tracking even in challenging environments, overcoming signal loss while entering buildings. Furthermore, the energy-efficient design enables accurate distance measurement, making this system a comprehensive solution for effective belongings management and retrieval in diverse settings.

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